



UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 30682

LM RELATIVISTIC ASTROPHYSICS

SUMMER EXAMINATIONS 2025

Time allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

Answer all four questions from this section.

1. Derive the formula for the Eddington luminosity limit of an object of mass M , and provide its value in Solar luminosity for $M = 10^6 M_\odot$. The Thomson scattering cross-section is $\sigma_T = 7 \times 10^{-25} \text{ cm}^2$. [10]

2. Due to magnetic braking, a pulsar loses rotational energy $E = I\Omega^2/2$ at a rate

$$\frac{dE}{dt} = -\frac{B_p^2 R^6 \Omega^4 \sin^2(\alpha)}{6c^3},$$

where $I = 2MR^2/5$ is the moment of inertia of the neutron star, Ω its angular frequency, M and R its mass and radius. The neutron star has a magnetic field of strength B_p at its pole, which is inclined by an angle α with respect to the rotational axis. Derive and compute a lower limit on B_p , using typical values for the neutron star's mass and radius, and a rotational period $P = 10^{-3} \text{ s}$ with time derivative $\dot{P} = 10^{-20}$. [10]

3. A star of mass $M_\star$ and radius $R_\star$ approaches a black hole of mass M_h . Derive an equation for the tidal radius, inside which the star gets tidally disrupted. Assuming that the star is a white dwarf, compute the maximum black hole mass, below which the fate of the disruption can lead to observable signatures. [10]

4. The mass loss rate from a black hole of mass M due to Hawking's radiation is given by

$$\frac{dM}{dt} = -\frac{\alpha \hbar}{M^2},$$

where $\alpha = c^4/(15360\pi G^2)$. Compute the timescale for a black hole to completely evaporate due to this effect. Primordial black holes are hypothetical objects that formed soon after the Big Bang with a large range of masses. Derive a lower limit on the mass of such black holes still present in our local Universe. [10]

SECTION 2

Answer **two** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

5. A core-collapse supernova explodes, launching a spherical ejecta of mass $M = 20M_{\odot}$ into space at a constant velocity $v = 10^4 \text{ km s}^{-1}$. Throughout this problem, you may assume that $t \gg t_0 = R_0/v$, where t is the time since the explosion and R_0 is the initial ejecta radius.

- (a) Write the dynamical timescale t_{dyn} required for the ejecta to reach a radius R . [2]
- (b) Describing the motion of photons inside the ejecta as a random walk through N successive scatterings, show that the diffusion timescale they take to reach the edge of the ejecta at a radius R can be written $t_{\text{diff}} = R\tau/c$, where $\tau = \rho\kappa R$ is the optical depth, ρ the density, and κ the opacity. [6]
- (c) Determine the time t_{lc} at which the supernova luminosity starts dropping as a function of M , v and κ , making use of the condition that the diffusing photons must “catch up” with the expanding ejecta. You may assume that the ejecta has a uniform density. Provide a numerical value for t_{lc} in days, using $\kappa = 1 \text{ cm}^2 \text{ g}^{-1}$. [6]
- (d) The ejecta has a radiation energy E , whose value at the time of explosion is $E_0 = 10^{51} \text{ erg}$. Neglecting energy loss due to photons leaving the ejecta, use the first law of thermodynamics to show that E decreases as the inverse of time. You may use that the radiation pressure is given by $P = E/(3V)$, where V is the volume of the ejecta. [6]
- (e) Evaluate the energy remaining in the ejecta at time t_{lc} and use it to estimate the peak supernova luminosity L_{peak} . Provide a numerical value for L_{peak} in erg s^{-1} for an initial ejecta radius $R_0 = 10R_{\odot}$. [6]
- (f) Is this luminosity high enough to account for that of a typical supernova? If not, provide details about the additional mechanism needed to boost the luminosity. [4]

ANY CALCULATOR

6. A circular supermassive black hole binary of mass $M = 10^6 M_\odot$ emits gravitational waves with a luminosity $L \approx G \ddot{I}^2 / c^5$, where I is the quadrupole moment of the binary and the dots denote time derivatives.
- (a) At the order-of-magnitude level, show that $L \approx c^5 (R_g/a)^5 / G$, where $R_g = GM/c^2$ and a is the binary separation. **[4]**
- (b) Assuming that supermassive black hole binaries form with a separation $a = 1$ pc, estimate the time needed for them to merge due to the energy lost through gravitational waves. Can this mechanism alone be responsible for the mergers detectable today? Justify your answer quantitatively. **[8]**
- (c) If the binary eventually merges, compute the frequency of the emitted gravitational wave signal at the onset of merger and provide its value in Hz. The merger is assumed to start when the binary separation reaches $a = 6R_g$, that is the radius of the innermost stable circular orbit. **[5]**
- (d) The LISA interferometer will detect gravitational waves at this frequency with a strain sensitivity threshold of $h_{\text{thr}} = 10^{-21}$. Compute the distance in Gpc within which this instrument will detect the signal from the merger with a strain a thousand times larger than the threshold. Make use of an order-of-magnitude calculation based on the quadrupole formula $h \approx G \ddot{I} / (c^4 d)$ for the strain, where d is the distance to the gravitational wave source. **[7]**
- (e) At the time of merger, gas is present between the two black holes. Assuming that it gets entirely accreted over the duration of the merger with a 10% radiative efficiency, compute the mass of gas needed for the resulting emission to reach the Eddington luminosity of the binary. **[6]**

ANY CALCULATOR

7. A white dwarf with mass $M_\star = 0.1M_\odot$ and radius $R_\star = 10^{-2}R_\odot$ is supported by the degeneracy pressure of electrons.

(a) Is the white dwarf stable? Justify your answer quantitatively. **[2]**

(b) A passing cloud accretes onto the white dwarf, leading to electromagnetic emission with luminosity $L = 10^{36} \text{ erg s}^{-1}$ for 10 years. Assuming that radiation is released by the gas hitting its surface, determine whether the white dwarf survives this accretion episode or explodes as a supernova. **[8]**

(c) The white dwarf gets captured by an intermediate-mass black hole of mass $M_h = 10^4 M_\odot$ on a parabolic orbit with a pericentre equal to the tidal radius, causing the star to be disrupted.

i. Compute the value of the tidal radius in AU. **[2]**

ii. Making use of the frozen-in approximation, determine the spread in orbital energy imparted to the stellar debris and describe qualitatively the evolution of this gas following the disruption. **[7]**

iii. Show that the orbital period of the most bound debris is about an hour. Assuming that a third of the white dwarf's mass is accreted over this timescale, determine the accretion rate onto the black hole. **[8]**

iv. For a 1% radiative efficiency, compute the resulting accretion luminosity in terms of the Eddington luminosity. How is the gas surrounding the black hole expected to evolve? Justify your answer qualitatively by comparing the different forces involved. **[3]**

Formula Sheet

All necessary formulae are provided as part of the questions.

Astrophysical Constants and Units

Astronomical Unit	AU	$1.50 \times 10^{11} \text{ m}$
Parsec	pc	$3.1 \times 10^{16} \text{ m}$
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_{\odot}$	$3.84 \times 10^{26} \text{ W}$
Absolute bolometric magnitude of the Sun	M_{bol}	$+4^{\text{m}}75$
Bolometric correction for the Sun	BC	$-0^{\text{m}}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\text{m}}74$
Solar constant	f	$1.36 \times 10^3 \text{ W m}^{-2}$
Solar mass	$M_{\odot}$	$1.989 \times 10^{30} \text{ kg}$
Solar radius	$R_{\odot}$	$6.95 \times 10^8 \text{ m}$
Solar effective temperature	T_{eff}	5777 K
Wien constant	b	$2.898 \times 10^{-3} \text{ m K}$
Hubble constant	H_o	$70 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_{\oplus}$	$5.972 \times 10^{24} \text{ kg}$
Earth radius	$R_{\oplus}$	6371 km
Jupiter mass	M_{Jup}	$1.898 \times 10^{27} \text{ kg}$
Jupiter radius	R_{Jup}	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.